

Impact of Global Crises on Financial Markets

DSA 210 — Introduction to Data Science · Spring 2026

Yağmur Çaloğlu · Sabancı University

GitHub: github.com/yagmurcaloglu-design/DSA_210_TermProject

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Abstract

This project tests whether six global crises between 2000 and 2024 produced measurable shifts in financial markets, and whether those shifts could be predicted. Daily price data for ten assets was combined with macroeconomic indicators to build a dataset of 6,292 observations. Three hypothesis tests (Mann-Whitney U, Kruskal-Wallis, Interrupted Time Series) and three machine learning methods (Logistic Regression, Random Forest, K-Means) were applied. ITS detected significant shifts in 5 of 6 crises, while the ML models performed at chance level ($AUC \approx 0.51$). The combined result supports the weak-form Efficient Market Hypothesis: crises move markets in detectable ways, but by the time they are publicly identifiable, the information is already absorbed into volatility and sector behavior.

1. Motivation

Global crises reshape financial markets, but the headlines rarely match what the data shows. "Markets crashed" is a useful summary for a newspaper, but sector behavior is usually more nuanced: defense rises in geopolitical events, aviation falls in pandemics, and gold behaves almost independently of the broader index. This project asks two questions:

- Do global crises produce statistically detectable shifts in market returns and sector behavior?
- If they do, can a model predict the next-day market direction using those shifts?

The motivation is twofold. First, the project applies the full data-science pipeline (collection, EDA, hypothesis testing, ML) to a domain known for fat-tailed, non-stationary returns. Second, it tests the Efficient Market Hypothesis in practice: if a model cannot predict markets even with crisis labels, that is itself evidence the market has already absorbed the information.

The project covers six crises spanning three categories:

- **Financial:** 2008 Global Financial Crisis
- **Health:** COVID-19 pandemic (March 2020)
- **Geopolitical:** 9/11 attacks (2001), Iraq War (2003), Russia-Ukraine invasion (2022), Israel-Hamas conflict (2023)

2. Data Source

Data was collected programmatically from two public sources:

- **yfinance (Yahoo Finance API)** — daily closing prices for S&P 500, VIX, WTI Crude Oil, Gold, and five sector ETFs (Defense, Energy, Healthcare, Technology, Aviation).
- **FRED (Federal Reserve Economic Data)** — Federal Funds Rate, CPI, Unemployment Rate, and M2 money supply. MSCI World was added as a global benchmark.

The aviation series required reconstruction: the JETS ETF only began trading in 2015 and produced 3,858 missing values. To get full historical coverage, I built a composite using Boeing (BA) for pre-2007 data and Delta Airlines (DAL) for post-2007.

Final dataset: **6,292 daily observations across 10 asset series, 2000-01-03 to 2024-12-31**. Returns were computed as daily percentage changes. Crisis windows were ± 30 trading days around each event for the main analysis, extended to ± 60 days for ITS.

3. Data Analysis

3.1 Preprocessing

After building the aviation composite, daily returns were computed for all assets and macroeconomic series were aligned to the trading calendar. Engineered features included:

- **is_crisis** — binary flag for days within 30 days of any crisis
- **SP500_ma5**, **SP500_ma20** — 5- and 20-day moving averages
- Per-crisis pre/post window slices (saved separately as CSV files)

3.2 Exploratory Data Analysis

Three patterns came out clearly:

- **Heavy tails everywhere** — daily returns are far from normally distributed, which ruled out the t-test as a primary tool and motivated non-parametric methods.
- **Sector responses differ by crisis type** — defense rises in geopolitical events (9/11: +1.03% daily), aviation collapses in health crises (COVID-19: −2.18%), energy swings either way depending on the cause.
- **Correlation structure is informative** — Tech sits at 0.88 with the S&P 500, while Gold is near 0.00, making gold the most effective diversifier in the dataset. VIX is at −0.71, confirming its role as a fear gauge.

3.3 Hypothesis Testing

Three tests were used, with non-parametric methods as the primary choice:

- **Mann-Whitney U** — compares pre vs post returns, no normality assumption.
- **Kruskal-Wallis** — compares all six crises at once.
- **Interrupted Time Series (ITS)** — regression that models the pre-crisis trend and measures both an immediate level change and a slope change after the event. Robust standard errors were used because the residuals showed strong autocorrelation (Durbin-Watson = 0.21–0.77).

MWU and Kruskal-Wallis found nothing significant at the 30-day window (Kruskal-Wallis $H = 5.49$, $p = 0.36$). This is a power problem, not a real null: with only ~20 trading days per side and very volatile daily returns, the tests cannot pick up real differences. ITS, which uses more information from the surrounding window, told a different story:

Crisis	β_2 (level)	β_2 p	β_3 (slope)	β_3 p	Sig?
9/11 (2001)	−9.98	0.0000	+0.347	0.0000	Yes
Iraq War (2003)	+4.69	0.0002	+0.437	0.0000	Yes
Financial Crisis (2008)	−7.24	0.0127	−0.483	0.0000	Yes
COVID-19 (2020)	−21.54	0.0000	+0.444	0.0000	Yes
Russia–Ukraine (2022)	+1.89	0.3735	+0.006	0.9297	No
Israel–Hamas (2023)	−2.61	0.1039	+0.326	0.0000	Yes

ITS found significant shifts in 5 of 6 crises. The exception — Russia–Ukraine — is informative on its own: markets had already priced in the geopolitical risk before February 2022, so the invasion did not produce a statistical break.

3.4 Machine Learning

The dataset was split chronologically: train = 2000–2020 (4,898 days), test = 2020–2024 (1,225 days). Class balance was reasonable (3,283 Up vs 2,840 Down). Three methods were applied:

- **Logistic Regression** — predicts whether the next day's S&P 500 return is positive, using lagged returns, VIX, sector returns, moving averages, and crisis features.
- **Random Forest Classifier** — same task, allowing non-linear interactions and producing a feature-importance ranking.

- **K-Means Clustering** — unsupervised regime detection. The elbow method picked $k = 3$, which I interpreted as Calm, Turbulent, and Crisis regimes.

4. Findings

4.1 The choice of test matters more than the result

Mann-Whitney U and Kruskal-Wallis returned no significant effects for any crisis. That looked like a clear null until I switched to ITS, which found significant level or slope changes in 5 of 6 crises. The difference is power: MWU compares two small samples, while ITS uses the full time series and explicitly models the pre-crisis trend. For event-study questions like this one, ITS is the more appropriate tool.

4.2 Markets are not easily predictable, even with crisis labels

- Logistic Regression: accuracy 0.54, AUC 0.516
- Random Forest: accuracy 0.53, AUC 0.507
- Both models predict Up days well (recall ≈ 0.86) but miss Down days (recall ≈ 0.15) — they essentially "bet on the market going up"
- `is_crisis` ranked **near zero** in Random Forest importance
- Top features were **Defense and Energy sector returns**, then S&P 500 moving averages and VIX

This is not a model failure — it is empirical support for the weak-form Efficient Market Hypothesis. By the time a crisis is publicly identifiable, the information has already been absorbed into volatility and sector behavior. A binary "crisis or not" label adds nothing on top of features the market is already pricing.

4.3 Regimes describe market state better than event dates

K-Means found three regimes from the asset return matrix:

- **Calm** — 4,383 days, average S&P 500 return +0.52%, VIX below average
- **Turbulent** — 1,739 days, average return −1.20%, VIX clearly elevated
- **Crisis** — a single day (April 20, 2020, when WTI oil futures briefly went negative, Oil = −306%)

The single-day Crisis cluster is an outlier, not a recurring market state. To get a more meaningful picture, I looked at the regime distribution in the ± 60 -day window around each crisis:

Crisis	% Turbulent	Pattern
Financial Crisis (2008)	43%	Turbulence clusters after crisis date (matches ITS negative slope)
Russia-Ukraine (2022)	36%	Turbulence elevated before invasion — already priced in
COVID-19 (2020)	31%	Sustained turbulence plus the unique oil-collapse Crisis day
9/11 (2001)	30%	Moderate, less persistent — typical geopolitical pattern
Iraq War (2003)	26%	Already partially anticipated
Israel-Hamas (2023)	24%	Lowest — markets priced as regional, not global

4.4 Sectors are more predictable than the aggregate

While the S&P 500's response is mixed across crisis types, sector behavior is surprisingly consistent:

- **Defense** — gains in geopolitical events (9/11: +1.03%, Russia-Ukraine: +0.77%)
- **Energy** — gains during war (Russia-Ukraine: +0.76%) but collapses in financial crises (2008: −1.75%)
- **Aviation** — worst hit in health crises (COVID-19: −2.18% daily)
- **Gold** — near-zero correlation with S&P 500 across the full sample, best diversifier in the dataset

4.5 US versus world

Comparing the S&P 500 against MSCI World (2010 onward): both indices moved together during COVID-19 (a truly global crisis), MSCI World was slightly more affected by Russia-Ukraine (higher European exposure), and the Israel-Hamas impact on both was minimal. Over the longer trend, the S&P 500 grew $\sim 5\times$ while MSCI World grew $\sim 2\times$ — a structural US outperformance that frames every crisis comparison.

5. Limitations and Future Work

Limitations

- The ML test period (2020–2024) includes COVID-19 and its aftermath, making the test set unusually volatile compared to the training set.
- Daily granularity misses intraday reactions, which often carry the strongest initial signal.
- Six crises is a small sample for generalizing about "crisis types."
- ITS assumes a linear pre-trend — a non-linear break would be misattributed to the event.
- The K-Means Crisis cluster collapsed to a single outlier day. This is why the ± 60 -day regime distribution analysis was added.

Future Work

- Use intraday or hourly data around event dates to catch initial reactions.
- Add news-sentiment features (headline tone, earnings call language) as an exogenous signal not already in prices.
- Try regime-switching models (Hidden Markov Models, Bayesian Online Changepoint Detection) that let regimes evolve over time instead of being hard-clustered.
- Test whether a sector-rotation strategy conditioned on the K-Means regime can outperform buy-and-hold S&P 500 net of transaction costs.

References

- Yahoo Finance API via the `yfinance` Python package
- Federal Reserve Economic Data (FRED), St. Louis Fed
- Course materials: DSA 210, Sabancı University, Spring 2026

AI Assistance Disclosure

This project was completed with the assistance of Claude (Anthropic, claude.ai). AI was used for:

- Coding help — debugging the JETS ETF NaN issue and constructing the BA+DAL composite aviation series.
- Methodological guidance — justifying Mann-Whitney U and ITS over the t-test given fat-tailed returns, and motivating robust standard errors from Durbin-Watson diagnostics.
- Structuring the ML notebook and motivating the ± 60 -day regime-distribution analysis after the K-Means "Crisis" cluster collapsed to a single outlier.
- Writing assistance for this report.

Full conversation logs are linked in the repository README. All analytical decisions, data interpretation, and final conclusions are the author's own.